DATA IDENTIFICATION, SOURCING STRATEGY AND INITIAL ANALYSIS PLANNING

Beauty & Wellness Industry

Business Problem: Predicting Customer Repeat Purchases

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Course / Module	Data Science – Week 2
Report Type	Data Sourcing Strategy and Initial Analysis Plan
Builds on	Week 1: Business Problem Definition and Proposed Analysis Plan

1. Executive Summary

This report sets out how data will be identified, assessed and collected to investigate the Week 1 problem: whether past purchasing behaviour can help identify customers who are more likely to make a repeat purchase of beauty and wellness products. It is a planning document. No dataset has been downloaded or analysed for this report, and no model results or business outcomes are claimed.

Ten candidate sources were reviewed, covering transaction and event logs, review data, product databases, a government safety dataset, search-trend data, a national open-data portal and social media. Each was judged on relevance, accessibility, reliability and time coverage, and its strengths and limits are listed in Sections 4 and 5. Where a source page could not be opened during this review, it is marked as not verified.

Recommended plan: use the REES46 cosmetics-shop event log (S1) and the Open e-commerce 1.0 purchase histories filtered to beauty items (S2) as the two main datasets, use the Olist marketplace data (S3) as a cross-market check, and use product, review and search-trend sources only for enrichment and context. Section 7 maps these sources to the three Week 1 hypotheses, Section 8 describes the exploratory analysis, Section 9 assesses data quality and relevance, and Section 10 lists the cleaning steps expected.

2. Recap of the Week 1 Problem

Problem statement: Can historical customer purchasing behaviour and transaction characteristics help identify customers who are more likely to make a repeat purchase of beauty or personal-care products within a defined future period?

ID	Week 1 research question or hypothesis
RQ1	How do purchase frequency and order value vary across customers?
RQ2	Are customers with more frequent past purchases more likely to purchase again in a defined future period?
RQ3	Do repeat-purchase patterns differ across beauty product categories?
RQ4	What data limitations could affect the reliability and usefulness of a prediction?
H1	Customers with more frequent past purchases may have a higher probability of a repeat purchase in the observation window.
H2	Repeat-purchase behaviour may vary across product categories.
H3	Time since a customer's last purchase may be associated with the likelihood of another purchase.

The Week 1 report listed a Kaggle cosmetics purchase-prediction page (https://www.kaggle.com/c/cosmetics/data) as a candidate. That page could not be opened in this review, so it is treated as unverified and the plan below does not depend on it.

3. Data Sourcing Strategy

3.1 What the problem needs

To test the hypotheses, a dataset needs at least: a pseudonymous customer identifier, a purchase date for each order, an order value or price, and a product category. Customer demographics, brand and promotion information are useful extras. The data must also cover enough time for a past period (to build predictors) and a later period (to observe the repeat-purchase outcome).

3.2 Selection criteria

Criterion	What is checked
Relevance	Does it contain customer-level purchases in beauty or personal care, so that a repeat-purchase label can be built?
Accessibility	Can it be downloaded or queried for free without special approval? Are the licence and permitted uses clear?
Reliability	Who collected it, is it documented, are known biases reported, is the structure consistent?
Time coverage / frequency	How long is the observation period and is the data updated? A longer span allows a more realistic prediction window.
Granularity	Order level or event level, and whether customers can be followed over time.
Ethics and privacy	Is the data pseudonymous, and do the terms prohibit re-identification?

3.3 Source tiers

- **Tier 1 – primary data:** customer-level purchase records used to build the repeat-purchase label (S1, S2).
- **Tier 2 – validation:** a different market or platform used to check whether patterns hold (S3).
- **Tier 3 – enrichment and context:** product, review, search-trend, safety and macro data that add background but cannot supply the label (S4–S9).
- **Not selected:** social media data, because it is not linked to purchases and carries privacy and terms-of-use risk (S10).

4. Candidate Data Sources

Table 1 gives an overview. Detailed notes, with links, strengths and limits, follow in Section 4.1. The last column states what was actually checked in this review.

ID	Source	Type	Verification in this review
S1	REES46 cosmetics shop events (Kaggle)	Event log	Details from a project README; Kaggle page not opened
S2	Open e-commerce 1.0 (Harvard Dataverse)	Purchase histories + survey	Article read; hosting, size, fields confirmed
S3	Olist Brazilian e-commerce (Kaggle)	Marketplace orders	Tables from a secondary summary; Kaggle page not opened

ID	Source	Type	Verification in this review
S4	Amazon Reviews 2023 (McAuley Lab)	Reviews + product metadata	Configuration names confirmed; terms not seen
S5	Sephora skin-care reviews (Kaggle)	Reviews	Not verified
S6	Open Beauty Facts	Product database / API	Licence and rate limits confirmed from docs
S7	openFDA cosmetic adverse events	Government safety reports	FDA and openFDA pages read
S8	Google Trends (API alpha, pytrends)	Search interest index	Announcement title only; details unverified
S9	data.gov.in	National open-data portal	Not verified; no beauty dataset identified
S10	Social media platforms	Posts and engagement	Not verified; not selected

Table 1. Candidate sources and what was verified.

4.1 Source profiles

S1 – REES46 – eCommerce events history in a cosmetics shop (Kaggle)

Role	Primary candidate (behavioural event log)
Link	https://www.kaggle.com/datasets/mkechinov/ecommerce-events-history-in-cosmetics-shop
Contents	Event-level log from an online cosmetics store: event_time, event_type (view, cart, remove_from_cart, purchase), product_id, category_id, category_code, brand, price, user_id, user_session. A public project using it reports about 9 million rows, roughly 713 thousand users and 45 thousand products for October–November 2019 (https://github.com/IIAndrewII/Customer_Behavior).
Access and licence	Free download with a Kaggle account. Licence and usage terms must be read on the dataset page before use.
Coverage / frequency	Historical snapshot, not updated. The number of months available must be confirmed on the Kaggle page (the project above used two months).
Strengths	Beauty-specific. A persistent user_id allows repeat purchases to be observed. Timestamps allow recency, frequency and inter-purchase gaps. Brand and category are included.
Limitations	Short time span limits the choice of prediction window. Very large share of view events. category_code is often missing. No demographics. Not India-specific. Bots or atypical users may be present.
Use in this project	Main dataset for H1, H2 and H3 if a repeat-purchase label can be built over the available months.
Verification status	Kaggle page could not be opened in this session. Fields and size are taken from a public project README. Confirm months, size and licence before relying on it.

S2 – Open e-commerce 1.0 – crowdsourced Amazon purchase histories

Role	Primary candidate (long purchase histories + demographics)
Link	https://doi.org/10.7910/DVN/YGLYDY Paper: https://www.nature.com/articles/s41597-024-03329-6 | Code and surveys: https://github.com/aberke/amazon-study
Contents	5,027 participants and 1,850,717 purchases from 2018 to 2022. amazon-purchases.csv holds order date, shipping state, price per unit, quantity, ASIN/ISBN, title and category. survey.csv holds demographics and shopping-related answers; fields.csv defines the fields.

Access and licence	Hosted on Harvard Dataverse. Published for research purposes only; the data may not be used to re-identify participants. A separate dataset licence was not stated on the article page, so check the Dataverse record.
Coverage / frequency	Five years of purchase history (2018–2022), collected between November 2022 and March 2023. Static.
Strengths	Long follow-up makes repeat purchase and inter-purchase gaps measurable. Product category and demographics are available. Peer-reviewed data descriptor with documented biases.
Limitations	General Amazon shopping, not beauty-only, so a beauty subset must be built from Category and Title. The authors report sample bias: ages 25–34 are 36.1% of the sample versus 17.4% in census data, 65+ are 3.1% versus 22.2%, and higher-income households are under-represented. 40,368 gift-card purchases are concentrated among a few users. Category values are sometimes inconsistent with Title or missing. US only.
Use in this project	Second main dataset, and the best one for testing recency and frequency over years (H1, H3) and for category comparison (H2) after filtering beauty items.
Verification status	Article page read in this session (hosting, size, fields and limitations confirmed). Beauty-related category values still need to be inspected in the file.

S3 – Olist – Brazilian e-commerce public dataset (Kaggle)

Role	Validation candidate (marketplace orders)
Link	https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
Contents	About 100,000 orders from 2016 to 2018, split into related tables: orders, order items, payments, products, customers and a category-name translation table. customer_unique_id identifies returning customers. The product categories include Health Beauty.
Access and licence	Free download with a Kaggle account. Licence must be checked on the dataset page.
Coverage / frequency	Historical (2016–2018), not updated.
Strengths	Multi-table structure supports realistic joins. Order status, payment value and delivery dates allow richer features. Beauty category is available for filtering.
Limitations	Brazilian marketplace data from several sellers. Older. Beauty subset is only part of the data. It is expected that most customers place a single order, which gives a strongly imbalanced label (to be confirmed in EDA).
Use in this project	Out-of-context check of whether findings from S1 and S2 hold in a different market, and a source of order-level features.
Verification status	Table and column names confirmed from a secondary project summary (https://mavenanalytics.io/project/5666). Kaggle page not opened. Confirm licence and category names.

S4 – Amazon Reviews 2023 – All_Beauty and Beauty_and_Personal_Care (McAuley Lab)

Role	Supporting (reviews and product metadata)
Link	https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023 Paper: https://arxiv.org/abs/2403.03952

Contents	Per-category review files and product metadata files in JSONL form. Configurations include raw_review_All_Beauty, raw_meta_All_Beauty, raw_review_Beauty_and_Personal_Care and raw_meta_Beauty_and_Personal_Care. The whole collection is in the 10B–100B size category on its dataset card.
Access and licence	Free via Hugging Face. Terms and licence were not visible in the page content retrieved; read the dataset card before use.
Coverage / frequency	Multi-year, very large. Exact span to be confirmed on the dataset card.
Strengths	Very large beauty-related volume. Reviewer IDs and timestamps allow repeat engagement to be studied. Product metadata and review text support category and sentiment features.
Limitations	Reviews are not purchases, so repeat reviewing is only a weak proxy for repeat buying. Large files need sampling. Reviewer behaviour is self-selected.
Use in this project	Enrich products with ratings and review sentiment; study language about repurchase intention. Not used as the repeat-purchase label.
Verification status	Dataset configuration names confirmed on Hugging Face. Fields, span and licence to be confirmed.

S5 – Sephora products and skincare reviews (Kaggle)

Role	Supporting (skin-care reviews)
Link	https://www.kaggle.com/datasets/nenamalikah/sephora-products-and-skincare-reviews
Contents	Reported to contain product information and customer skin-care reviews, including reviewer attributes such as skin type.
Access and licence	Free download with a Kaggle account. Licence to be checked.
Coverage / frequency	Historical snapshot.
Strengths	Skin-care focus. Reviewer attributes could help describe customer groups.
Limitations	Review data, not purchase data. One retailer, so brand bias is likely. Coverage and fields not verified.
Use in this project	Optional descriptive context for product categories and customer attributes.
Verification status	NOT VERIFIED. This listing could not be checked in this session. Confirm that it exists, its fields and its licence before including it in the final plan.

S6 – Open Beauty Facts (crowdsourced product database and API)

Role	Enrichment (product attributes)
Link	https://world.openbeautyfacts.org API documentation (Product Opener): https://openfoodfacts.github.io/openfoodfacts-documentation/docs/Product-Opener
Contents	Open database of cosmetics and personal-care products: barcode, name, brand, categories and other product information, searchable through an API.
Access and licence	Read operations are public. Writing requires authentication. Data is described as available under the Open Database License. Rate limiting applies, so requests should be cached. API v3 is the current version.
Coverage / frequency	Continuously updated by contributors.

Strengths	Free and open. Gives brand and category information to clean and standardise product names from other datasets.
Limitations	Crowdsourced, so coverage is uneven and fields are often incomplete. No customers, sales or prices.
Use in this project	Standardise brand and category labels and add product attributes where barcodes or names match.
Verification status	Licence, authentication and rate-limit statements confirmed from the Product Opener documentation. Endpoints and download options to be checked in the project documentation.

S7 – FDA / openFDA – cosmetic adverse event reports

Role	Contextual (safety and trust)
Link	https://open.fda.gov/apis/cosmetic/event FDA announcement: https://www.fda.gov/food/cfsan-constituent-updates/fda-begins-posting-adverse-event-report-data-foods-and-cosmetics
Contents	Reports of cosmetic adverse events, downloadable as CSV and available through the openFDA API. At launch the FDA page cited 4,322 cosmetic reports from January 2004 to September 2016.
Access and licence	Public. openFDA terms of service apply; the page warns that access may be restricted under those terms.
Coverage / frequency	openFDA lists coverage from 2001 and an update frequency of 'infrequently'.
Strengths	Government source. Useful to discuss product safety and trust, which may influence repurchase.
Limitations	Reports are mostly voluntary and are not proof that a product caused the event. They include non-medical complaints. No purchase or customer information. US only.
Use in this project	Background only. It cannot be linked to purchase behaviour and will not be used in modelling.
Verification status	openFDA and FDA pages read in this session.

S8 – Google Trends (API alpha and pytrends)

Role	Contextual (demand and seasonality)
Link	https://developers.google.com/search/blog/2025/07/trends-api Unofficial Python library: https://pypi.org/project/pytrends/
Contents	Relative search interest over time for terms such as skincare or sunscreen, by region.
Access and licence	Google announced a Trends API in alpha in July 2025; access conditions were not available in the text retrieved. pytrends is an unofficial library and can break when Google changes its site.
Coverage / frequency	Frequent (regularly updated). Values are relative indices, not search counts.
Strengths	Free, frequent and can be limited to India. Useful for seasonality and campaign timing.
Limitations	Relative values only. Alpha access may be limited. Unofficial scraping may be unstable. Not linked to individual customers.
Use in this project	Check whether purchase timing in S1 and S2 follows seasonal interest (supporting H3).
Verification status	Announcement title and date confirmed. Details of access to be checked on the Google page.

S9 – Open Government Data Platform India (data.gov.in)

Role	Contextual (macro data for India)
Link	https://data.gov.in
Contents	India's national open-data portal. This search did not identify a beauty-specific customer or purchase dataset. It may hold retail, price-index or consumption statistics useful as background for the Indian market.
Access and licence	Public portal. Individual datasets have their own terms.
Coverage / frequency	Varies by dataset.
Strengths	Official source and relevant to the Indian context that Week 1 identified as a limitation.
Limitations	No confirmed beauty-specific dataset. Aggregated data cannot support customer-level prediction.
Use in this project	Macro context and a limitation discussion. A keyword search of the portal is planned for Week 3.
Verification status	Portal page could not be opened in this session; content is not verified.

S10 – Social media platforms (for example Reddit, Instagram, YouTube)

Role	Not selected (qualitative context only)
Link	No dataset link. Platform developer terms must be checked first.
Contents	Posts, comments and engagement about beauty products, which could be used for sentiment and topic analysis.
Access and licence	Access rules, API limits and permitted uses differ by platform and change over time. Personal data is involved.
Coverage / frequency	Continuous, but access to historical data is often restricted.
Strengths	Shows how customers talk about products and what concerns them.
Limitations	Not linked to purchases. Self-selected and possibly promotional content. Privacy and terms-of-use risk. Representativeness is unknown.
Use in this project	Not used for modelling. May inform qualitative understanding only if terms allow.
Verification status	Not verified. No specific dataset was selected.

5. Comparative Assessment of Sources

Each source was scored from 1 (weak) to 5 (strong) on four criteria for the specific purpose of predicting repeat purchases. The scores are the author's judgement based on the information gathered in Section 4; they are not measured values. Relevance is counted twice in the weighted score because a source that cannot supply a repeat-purchase label is of limited use however easy it is to access.

ID	Source	Relevance	Accessibility	Reliability	Time coverage	Weighted score
S1	REES46 – eCommerce events history in a cosmetics shop	5	4	3	2	3.8
S2	Open e-commerce 1.0 – crowdsourced Amazon purchase histories	4	4	3	4	3.8

ID	Source	Relevance	Accessibility	Reliability	Time coverage	Weighted score
S4	Amazon Reviews 2023 – All_Beauty and Beauty_and_Personal_Care	2	4	4	5	3.4
S3	Olist – Brazilian e-commerce public dataset	3	4	4	2	3.2
S6	Open Beauty Facts	2	5	3	4	3.2
S8	Google Trends	2	3	3	5	3.0
S5	Sephora products and skincare reviews	2	4	3	3	2.8
S9	Open Government Data Platform India	1	4	4	3	2.6
S7	FDA / openFDA – cosmetic adverse event reports	1	4	3	3	2.4
S10	Social media platforms	2	2	2	4	2.4

Table 2. Scores from 1 to 5; weighted score = (2 × relevance + accessibility + reliability + time coverage) / 5.

S1 and S2 rank highest, which matches the plan to use them as the main datasets. S2 scores well on time coverage and demographics but is not beauty-only. S1 is the most relevant but has the shortest span. These two weaknesses are opposite, which is why both are used.

6. Data Collection Plan

6.1 Phases

Phase	Actions	Output
A. Access and terms	Open each Tier 1 and Tier 2 page. Record licence, permitted use, file list and size. Confirm the beauty-specific fields exist.	Source log (date checked, licence, version)
B. Download	Download S1, S2 and S3 into a raw-data folder that is never edited. Record file names, sizes and SHA-256 hashes.	Immutable raw data and checksums
C. Profile	Load a sample of each file. Count rows, customers, date range, missing values and category values.	Profiling table per dataset
D. Decide	Confirm that a repeat-purchase label with a reasonable window can be built. If not, choose the next-best source.	Final dataset decision and note
E. Enrich (optional)	Query Open Beauty Facts for matching products. Pull Google Trends series for a few beauty terms in India if access allows.	Product attributes and a trend table

6.2 API collection rules

- Cache every API response locally so a request is never repeated unnecessarily (Open Beauty Facts states that rate limits apply).
- Store the query, date and API version next to each response so results can be reproduced.
- Treat unofficial tools such as pytrends as best-effort. If they stop working, the project does not depend on them.

6.3 Folder and governance rules

- **data/raw** – original files, never edited. **data/interim** – cleaned intermediate files. **data/processed** – analysis-ready tables.
- Raw data is not committed to GitHub; only a README with download instructions and the source log is committed.
- Use only pseudonymous identifiers, do not try to identify individuals and do not link customers across datasets.
- Do not mix prices across datasets without converting currency and inflation; analyse each dataset separately first.

7. Using the Data to Test the Week 1 Hypotheses

ID	Needed variables	Source datasets	Method	Main visual
H1	Customer ID, purchase dates, count of orders before the reference date, repeat label after it	S1; S2 (beauty subset); S3 for checking	Compare past frequency between repeaters and non-repeaters (Mann–Whitney U); logistic regression with frequency as predictor; ROC-AUC against a baseline	Boxplot of past orders by outcome; repeat rate by frequency band
H2	Product category, repeat label	S1 category_code; S2 Category plus title keywords; S3 product category	Chi-square test of independence with Cramér's V; correction for multiple comparisons; Wilson confidence intervals	Bar chart of repeat rate by category with intervals
H3	Days since last purchase at reference date, repeat label, time to next purchase	S1; S2; S3	Survival (Kaplan–Meier) analysis of time to next purchase; logistic regression with recency; Mann–Whitney U	Histogram of gaps between purchases; survival curve
RQ1	Orders per customer, order value	S1; S2; S3	Descriptive statistics; skewness; percentiles	Histograms (log scale), Pareto-style cumulative curve
RQ4	Missingness, duplicates, coverage, bias indicators	All selected sources	Data-quality report per dataset	Missingness heatmap; coverage by month

Results will be reported as associations. The data cannot show that frequency, category or recency cause repeat purchase, and a campaign effect would need a controlled experiment, as stated in Week 1.

8. Initial Exploratory Analysis Plan

8.1 Defining the target without leakage

- Choose a reference date T. Build all predictors only from data before T.
- Repeat purchase label = at least one purchase in the window (T, T + W]. Start with a window W that fits the data span (for example 30 or 60 days for S1; longer for S2) and run a sensitivity check with a second window.
- Customers whose follow-up window runs past the end of the data are excluded (right censoring).

- Use a time-based split: earlier reference dates for training, later ones for testing.

8.2 Key variables

Variable	Definition	Source fields (as expected)
past_orders	Number of purchases before T	S1: purchase events per user_id; S2: rows per ResponseID; S3: orders per customer_unique_id
recency_days	Days from last purchase to T	event_time; Order date; order purchase timestamp
total_spend / avg_order	Sum and mean of order value before T	price; Purchase price per unit × Quantity; payment value
category	Beauty product category	category_code; Category; product category name
brand	Brand (where recorded)	S1 brand; S4/S6 metadata
repeat_label	1 if purchase in (T, T + W], else 0	Derived
demographics (optional)	Age band, state, income band	S2 survey.csv only

8.3 Descriptive statistics

- Counts of customers, orders, products and categories; date range; share of customers with one order versus more.
- Mean, median, standard deviation, percentiles and skewness for order value and frequency.
- Repeat rate overall and by category, brand, month and frequency band.
- Class balance of the label, because a rare outcome affects the choice of metrics.

8.4 Visualisation plan

Chart	Shows	Variables
Histogram (log scale)	Distribution of orders per customer and order value	past_orders, order value
Boxplot / violin	Past frequency and recency by repeat outcome	past_orders, recency_days, repeat_label
Bar chart with intervals	Repeat rate by category and brand	category, repeat_label
Line chart	Orders and repeat rate over time; seasonality	date, orders
Survival curve	Share of customers not yet repurchased over time	time to next purchase
Heatmap	Missing values by field and dataset	all fields
Correlation matrix	Relationships between numeric predictors (Spearman)	numeric features

8.5 Statistical methods

- **Group comparisons:** Mann–Whitney U for skewed numeric variables; chi-square test for category versus outcome with Cramér's V as effect size.
- **Correlation:** Spearman rank correlation, because frequency and spend are right-skewed.

- **Time-to-event:** Kaplan–Meier curves for time to next purchase, with censoring handled explicitly.
- **Baseline and model (only if the label is reliable):** majority-class baseline, then logistic regression; evaluate with precision, recall, F1 and ROC-AUC on the later time period.
- **Reporting:** confidence intervals and effect sizes alongside p-values, and a correction when many categories are compared.

9. Data Quality and Relevance Assessment

The ratings below are expectations based on the source documentation. They will be replaced by measured values once the files are profiled in Week 3.

Source	Expected quality issues	Relevance for repeat-purchase prediction	Overall
S1 REES46	category_code often missing; many view events; possible bots; short period; no demographics	High: beauty-specific with customer IDs and timestamps; the short span restricts the prediction window	High, with a window limit
S2 Open e-commerce	Sample skews young and higher-educated; gift cards inflate counts; category sometimes inconsistent or missing; US only	High for frequency and recency; beauty subset must be built; limited generalisation	Medium–High
S3 Olist	Older (2016–2018); marketplace; expected strong class imbalance; beauty subset only	Medium: a useful cross-market check	Medium
S4 Amazon Reviews	Reviews are not purchases; very large; self-selected reviewers	Low–Medium: enrichment only	Low–Medium
S5 Sephora reviews	Not verified; review data; one retailer	Low	Low (unverified)
S6 Open Beauty Facts	Crowdsourced; uneven completeness	Low: product attributes only	Low–Medium
S7 openFDA	Voluntary reports; no causation; includes non-medical complaints	Very low: no purchase link	Low
S8 Google Trends	Relative index; alpha or unofficial access	Low–Medium: seasonality context	Medium
S9 data.gov.in	No beauty dataset identified; aggregated data	Very low: macro background	Low

9.1 Fitness for the Week 1 problem

- **Can the label be built?** Yes in principle for S1, S2 and S3, because each follows customers over time. The span in S1 and the sample size of S2 will decide the window.
- **Is the beauty focus preserved?** S1 yes; S2 only after filtering; S3 only for its beauty categories.
- **Does it represent Indian customers?** No selected dataset is from India. Any conclusion has to be stated as applying to the dataset's market, as Week 1 required.
- **Could the data answer RQ4?** Yes. The profiling step itself produces the evidence on missingness, bias and coverage.

10. Data Cleaning and Preprocessing Plan

Issue	Planned action	Check
Duplicate records	Remove exact duplicate order rows; keep event logs as recorded but flag repeated identical events	Count before and after

Issue	Planned action	Check
Missing category (S1, S2)	Map category_id or title keywords to a category where possible; otherwise label 'unknown'. Do not guess values	Share of unknown reported
Event types (S1)	Use purchase events for the label; keep view and cart events only as optional features	Event-type counts
Dates and time zones	Parse to datetime; convert to a single time zone; remove impossible or out-of-range dates	Min and max date per file
Price and quantity outliers	Inspect zero, negative and extreme values; use log scale for plots; remove only clear errors and record why	Before and after summary
Gift cards (S2)	Exclude gift-card purchases from frequency counts	Count removed
Inconsistent category and title (S2)	Build a beauty filter from category plus title keywords; manually review a random sample	Sample accuracy noted
Cancelled or unavailable orders (S3)	Keep completed orders only; document the rule	Order status counts
Text fields	Lowercase, trim and standardise brand and category names	Number of unique values before and after
Atypical users	Flag users with extreme event counts as possible bots; test results with and without them	Sensitivity result
First-observed purchase	Treat the first observed order as not necessarily the first ever; note left censoring	Stated in limitations
Class imbalance	Report label balance; choose metrics to match; avoid accuracy alone	Label distribution table
Privacy	Keep identifiers pseudonymous; drop unneeded personal fields; no cross-dataset linking	Field list reviewed

11. Risks, Ethics and Limitations

- **Source verification:** several pages could not be opened during this review. Licence, size and field details must be confirmed before any dataset is used, and the report must not claim use of a dataset that has not been obtained.
- **Short or biased samples:** S1 covers a limited period and S2 over-represents some groups. Results may not generalise.
- **Market mismatch:** no selected dataset comes from India.
- **Association only:** the analysis cannot establish causation.
- **Terms of use:** S2 is for research use and forbids re-identification. Kaggle and Hugging Face datasets have their own terms.
- **Reproducibility:** a source log, checksums and a documented cleaning log will be kept.

12. Next Steps

- Open and confirm the licence and contents of S1, S2 and S3; update the source log.
- Download and profile the files; replace the expected quality ratings with measured ones.
- Fix the reference date and window; compute the repeat label and its balance.

- Start the EDA notebook and the first set of charts.
- Search data.gov.in for Indian retail or consumption statistics to support the market limitation.

13. References and Source Links

1. REES46 / Kaggle – eCommerce events history in cosmetics shop: https://www.kaggle.com/datasets/mkechinov/ecommerce-events-history-in-cosmetics-shop
2. Public project using the cosmetics events data: https://github.com/IIAndrewII/Customer_Behavior
3. Berke, A., Calacci, D., Mahari, R., Yabe, T., Larson, K., & Pentland, A. (2024). Open e-commerce 1.0, five years of crowdsourced U.S. Amazon purchase histories with user demographics. Scientific Data, 11, 491. https://www.nature.com/articles/s41597-024-03329-6
4. Open e-commerce 1.0 dataset (Harvard Dataverse): https://doi.org/10.7910/DVN/YGLYDY
5. Code and survey materials: https://github.com/aberke/amazon-study
6. Olist – Brazilian E-Commerce Public Dataset (Kaggle): https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
7. Amazon Reviews 2023 (McAuley Lab, Hugging Face): https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023
8. Amazon Reviews 2023 paper: https://arxiv.org/abs/2403.03952
9. Sephora products and skincare reviews (Kaggle, not verified): https://www.kaggle.com/datasets/nenamalikah/sephora-products-and-skincare-reviews
10. Open Beauty Facts: https://world.openbeautyfacts.org
11. Product Opener API documentation: https://openfoodfacts.github.io/openfoodfacts-documentation/docs/Product-Opener
12. openFDA – Cosmetic adverse events: https://open.fda.gov/apis/cosmetic/event
13. FDA – Adverse event report data for foods and cosmetics: https://www.fda.gov/food/cfsan-constituent-updates/fda-begins-posting-adverse-event-report-data-foods-and-cosmetics
14. Google Search Central – Google Trends API (alpha): https://developers.google.com/search/blog/2025/07/trends-api
15. pytrends (unofficial): https://pypi.org/project/pytrends/
16. Open Government Data Platform India: https://data.gov.in
17. Factors affecting consumers' repurchase intentions toward skin care cosmetics: A cross-sectional study in Vietnam. Heliyon (2024). https://doi.org/10.1016/j.heliyon.2024.e32285

Appendix A – Submission Description (200+ words)

This report presents a data identification and sourcing strategy for the beauty and wellness problem defined in Week 1: understanding whether past purchasing behaviour can help identify customers who are more likely to buy again. It reviews ten candidate public sources, including a cosmetics-shop event log, five-year crowdsourced purchase histories, a Brazilian marketplace dataset, product review collections, a crowdsourced product database, a government adverse-event dataset, search-trend data, India's open-data portal and social media platforms.

Each source is described with its link, contents, access conditions, time coverage, strengths and limitations, and is scored on relevance, accessibility, reliability and time coverage. Pages that could not be opened during the review are clearly marked as not verified. The report recommends two main datasets for building a repeat-purchase label, one cross-market validation dataset, and several enrichment and context sources.

The report explains how the data can test the three Week 1 hypotheses on purchase frequency, product category and recency, and sets out an initial exploratory analysis plan covering the target definition, key

variables, descriptive statistics, visualisations and statistical methods such as Mann–Whitney tests, chi-square tests and survival analysis. It also assesses the expected data quality and relevance of each source and lists the cleaning steps required, including handling missing categories, duplicates, outliers, gift cards and class imbalance.

This is a planning report. No dataset has been downloaded or analysed, and no model results or business outcomes are claimed. The next phase is to confirm the licence and contents of the chosen datasets, profile them, and begin exploratory analysis.

GitHub project URL: Add your repository URL here after creating and uploading the project. Do not enter a made-up URL.

Appendix B – Checklist Before Submission

- Confirm the student name and course details.
- Open each recommended source page and confirm the licence, files and size; update any 'not verified' entries.
- Create a GitHub repository and upload the project folder, including this report and the source catalogue.
- Upload the .docx file in the report upload field.
- Paste the submission description into the description box.
- Paste the real GitHub URL if the portal asks for it.
- Submit only after checking that the uploaded file is the final version.